

Creating Airfoil profiles from CSV files of Airfoil coordinates.

Important note, the CSV must be formatted correctly in order to be accepted by the custom block. The CSV entries must be of the form (x,y,z) and there must be no repeated points or points that are extremely close to each other (0.1 mm tolerance is sufficient). There also cannot be any other data inside of the CSV aside from the coordinates of the airfoil points.

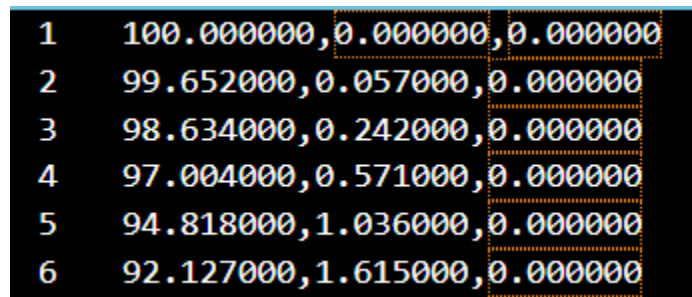
Block Overview

The CSV Airfoil SDF block creates an SDF of an airfoil profile given an input CSV. The block takes in a CSV file that contains the coordinates of the points of the airfoil. The CSV file must be formatted with the coordinates having the form of (x,y,z). Typical airfoil plotters such as [Airfoiltools.com](https://airfoiltools.com) export their coordinates as being (x,y) and with other metadata related to the airfoil which will cause issues when importing into nTop, therefore it is important to ensure the CSV file is in the correct format prior to importing. The conversion can be trivially performed by LLMs or simple Python script. The airfoil SDF can then be used to create wing planforms with the airfoil profile shape. The block takes in two inputs, the root point of the airfoil and the CSV file with the appropriate units. The output is the SDF of the airfoil profile.

Usage Steps

Step 1: Importing the CSV

The importing of the CSV is critical for the block to function properly. The CSV should be formatted with the points being of the form of (x,y,z) and there can be no overlapping points or points within 0.1 mm of each other so make sure that the CSV you intend to use is formatted correctly. The CSV contents should look similar to the ones in the image.

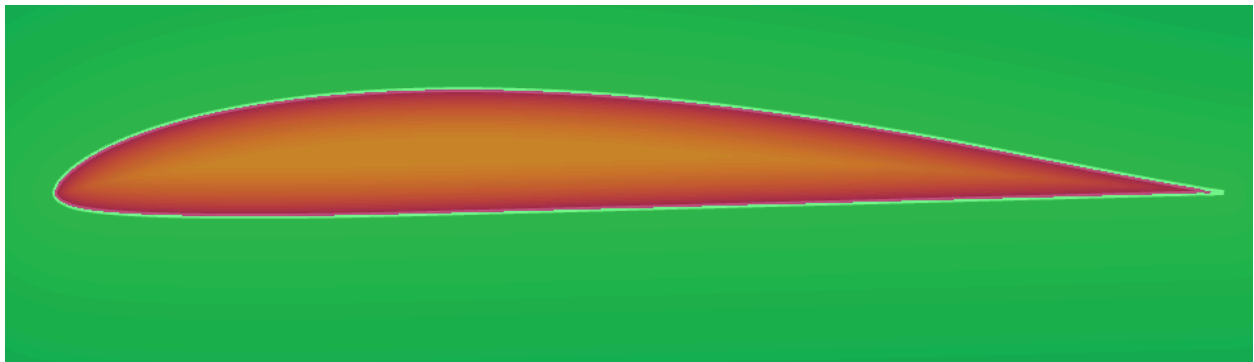


1	100.000000,	0.000000,	0.000000
2	99.652000,	0.057000,	0.000000
3	98.634000,	0.242000,	0.000000
4	97.004000,	0.571000,	0.000000
5	94.818000,	1.036000,	0.000000
6	92.127000,	1.615000,	0.000000

Once the CSV is in the correct format, the custom block simply takes in the file path of the CSV and the units of the coordinates in the CSV.

Step 2: Verifying your output

Once the CSV is imported the custom block should create the SDF of the airfoil profile automatically. It is critical that the field is verified to be correct since any errors in the field can lead to issues downstream when the airfoil profile is used downstream by other blocks or notebooks. To verify the output field, enter the field viewer and make sure that the field is enclosed within the profile of the airfoil. Common issues that can arise are points that are extremely close to each other but not being the exact same point. This issue can create an infinite field most commonly at the trailing edge of the airfoil since these points essentially make the profile loop back onto itself thus creating an infinite field outside of the profile. The fix to this issue is to ensure that there are no points within 0.1 mm of the first point of the CSV. This can be specified to an LLM or hardcoded into a script. The final result of the field should be look like the one in the image.



The important parts to notice of the field are that they are entirely contained within the airfoil profile and the profile smoothly converges at both the leading and trailing edges.

Notebook walkthrough

The custom block creates the airfoil profile by using two simple blocks. The CSV import creates the point list for the spline through points. By default the degree is set to 3 but a degree of 2 can also be useful for some airfoils. The spline through points block also has periodic on by default in order to create the closed shape of the airfoil. The spline through points result is then passed to a profile from curves block with the curves list only being occupied by the airfoil points spline. This creates the SDF for the airfoil which can then be used to create wing planforms.

Conclusion

In closing, this custom block takes in a CSV and outputs an airfoil profile as an SDF. This allows virtually all airfoil profiles that can have their profiles described by CSVs available to be used within nTop. This SDF can then be used to create wing planforms.